Miguel Peidro Paredes.
AI Security Architect



Security architect for generative AI at BBVA Corporate & Investment Banking since the bank's first AI project: security-by-design, risk assessment and audit of AI systems on AWS and Azure. Previously researched adversarial generative models against fingerprint authentication at UC3M (first author, ACM ARES 2023). I also build and maintain my own agent platform.

Madrid · EU citizen | miguelpeidroparedes@gmail.com | +34 675 758 812 | in/miguelpeidroparedes

miguelpeidrotrabajo.github.io/portfolio

100,000 h / yr
Hours saved yearly by automations I built at BBVA CIB.

245 +
Tested capabilities in my own agent platform.

ARES '23
First-author paper (ACM) on GAN-generated fingerprints.

$16M
Peak market cap of DFSocial, co-founded at 21.

Experience

AI Security Architect & IT Security Auditor
2024–now
BBVA · Corporate & Investment Banking, via Nfoque Advisory Services

- Security architect for generative AI at BBVA CIB since the bank's first AI project: security-by-design, risk assessment and review of AI architectures on AWS and Azure.
- Security audits and mitigation plans for critical banking systems under DORA, NIST and ISO 27001.
- Built internal automations that save an estimated 100,000 hours of work per year across CIB teams.

AI Researcher & Engineer · Guest Lecturer
2022–2024
Universidad Carlos III de Madrid

- Generative adversarial networks that synthesise fingerprint patterns (MasterPrints), tested against Android capacitive sensors. First author, ACM ARES 2023.
- Designed and delivered AI-security training for INCIBE and the Spanish National Police.

Cybersecurity & Distributed Systems Analyst
2021–2023
FUNDITEC

- Engineered secure distributed architectures and smart contracts for Horizon 2020 and NextGenerationEU projects (supply-chain and renewable-energy traceability, Gaia-X, Peppol).

Co-founder & IT Security Manager
2020–2021
DFSocial Gaming · DecodedWorld

- Co-founder and head of IT security of DFSocial, the first DeFi gaming ecosystem (Ethereum, BNB Smart Chain). Peak market cap of $16M; team of 16.

Research

Characterizing the MasterPrint threat on Android devices with capacitive sensors — 2023
First author. ARES 2023, ACM · doi.org/10.1145/3600160.3600177

Doctoral research, AI applied to cybersecurity · UC3M, INCIBE-supported; paused in 2024 — 2022–2024

Project

Hybrid Agentic Platform — Independent · ongoing
Cloud and local agent platform for companies, built to lower deployment cost. 245+ tested capabilities.

Education

MSc Cybersecurity, UC3M · thesis: APT analytics with geopolitical data (GDELT) — 2020–2021

BSc Computer Engineering, UC3M · thesis: software for malicious macro detection — 2016–2020

Learning
2025–2026

CMU 11-768 · AI Agents	in progress	Berkeley · Agentic AI	done
AI Security Bootcamp (AISB)	done	BlueDot · Technical AI Safety	done
Stanford CS224W · ML with Graphs	done	Stanford · Cryptography I	done
UIUC · Cloud Computing Concepts 1–2	done	Helsinki · Big Data Platforms	done

Skills

Security	Security-by-design for GenAI & agents · Threat modelling · Adversarial ML · DORA · NIST · ISO 27001
AI & agents	LLM agents · Multi-agent systems · Hybrid cloud/local inference · RAG · GANs · Process automation
Platforms	AWS · Azure · Ethereum / BNB Chain · Smart contracts
Languages	Spanish (native) · English (C1) · French (A2)

Full experience, research and coursework at **miguelpeidrotrabajo.github.io/portfolio**

Scan for portfolio